# Practical 4.3: CRD REST API — Mwanza Statistics Dashboard

Consume CRD API and WFS in Python to produce a statistics dashboard

Day 4 · 3 hours · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

**Learning Objectives:**

- Search and list datasets on the CRD using the REST API
- Fetch Mwanza business data programmatically with Python
- Build a simple statistics dashboard with Pandas and Matplotlib
- Export analysis results for reporting

## Exploring the CRD REST API

```
# API base URL
CRD="https://crd.resilienceacademy.ac.tz"

# List all datasets (first page)
curl -s "$CRD/api/v2/datasets/?page_size=5" | python3 -m json.tool | head -40

# Search for Mwanza datasets
curl -s "$CRD/api/v2/datasets/?search=mwanza" | \
  python3 -c "import sys,json; d=json.load(sys.stdin); \
  [print(r['title']) for r in d['resources']]"

# Search for Tanga datasets
curl -s "$CRD/api/v2/datasets/?search=tanga" | \
  python3 -c "import sys,json; d=json.load(sys.stdin); \
  [print(r['title']) for r in d['resources']]"
```

## Mwanza Statistics Dashboard — Python

Save as `/sdi-training/crd_dashboard.py`:

```
#!/usr/bin/env python3
# CRD Statistics Dashboard - Mwanza and Tanga Data
import requests
import geopandas as gpd
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
import json

CRD = "https://crd.resilienceacademy.ac.tz"
print("Connecting to Resilience Academy CRD...")

#  1. List Mwanza datasets via REST API
r = requests.get(f"{CRD}/api/v2/datasets/?search=mwanza&page_size=20")
mwanza_datasets = r.json()
print(f"\nMwanza datasets found: {mwanza_datasets['total']}")
for ds in mwanza_datasets['resources']:
```

```
    print(f" - {ds['title']} ({ds['subtype']})")

#  2. Load Mwanza businesses via WFS
print("\nLoading Mwanza businesses via WFS...")
wfs_url = (f"{CRD}/geoserver/ows?service=WFS&version=2.0.0"
          "&request=GetFeature"
          "&typeName=geonode:mwanza_businesses"
          "&outputFormat=application/json")
gdf = gpd.read_file(wfs_url)
print(f"Loaded {len(gdf)} Mwanza business features")
print(f"Columns: {list(gdf.columns)}")

#  3. Statistics
print("\n--- Mwanza Business Statistics ---")
print(f"Total businesses: {len(gdf)}")
print(f"Bounding box: {gdf.total_bounds}")

# Get business type column (common field names in OSM data)
type_col = None
for col in ['amenity','shop','office','landuse','building']:
    if col in gdf.columns:
        type_col = col
        break

if type_col:
    counts = gdf[type_col].value_counts().dropna()
    print(f"\nTop 10 by {type_col}:")
    print(counts.head(10).to_string())

#  4. Build Dashboard Plot
fig = plt.figure(figsize=(16, 10))
fig.suptitle('Mwanza Businesses  CRD Data Analysis\n'
            'Source: crd.resilienceacademy.ac.tz',
            fontsize=14, fontweight='bold', color='#004640')

gs = gridspec.GridSpec(2, 2, figure=fig, hspace=0.35, wspace=0.3)

# Map
ax1 = fig.add_subplot(gs[:, 0])
gdf.plot(ax=ax1, color='#007064', markersize=5, alpha=0.6)
ax1.set_title('Mwanza Business Locations', fontweight='bold', color='#004640')
ax1.set_xlabel('Longitude'); ax1.set_ylabel('Latitude')
ax1.grid(True, alpha=0.3)

# Bar chart by type
if type_col and len(counts) > 0:
    ax2 = fig.add_subplot(gs[0, 1])
    colors = ['#007064' if i == 0 else '#A8DDD8' for i in range(min(10, len(counts)))]
    counts.head(10).plot(kind='barh', ax=ax2, color=colors)
    ax2.set_title(f'Businesses by {type_col.title()}', fontweight='bold', color='#004640'
        )
    ax2.set_xlabel('Count')
    ax2.invert_yaxis()

# Summary stats
ax3 = fig.add_subplot(gs[1, 1])
ax3.axis('off')
summary = [
    ['Metric', 'Value'],
```

```
    ['Total businesses', str(len(gdf))],
    ['Geometry type', str(gdf.geom_type.unique()[0])],
    ['CRS', str(gdf.crs)],
    ['Data source', 'crd.resilienceacademy.ac.tz'],
    ['City', 'Mwanza, Tanzania'],
    ['Layer', 'geonode:mwanza_businesses'],
]
table = ax3.table(cellText=summary[1:], colLabels=summary[0],
                 cellLoc='left', loc='center')
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1.2, 1.8)
for (r, c), cell in table.get_celld().items():
    if r == 0:
        cell.set_facecolor('#007064')
        cell.set_text_props(color='white', fontweight='bold')
    else:
        cell.set_facecolor('#D5F0EC' if r % 2 == 0 else 'white')

plt.savefig('mwanza_crd_dashboard.png', dpi=150, bbox_inches='tight',
           facecolor='white')
print("\nDashboard saved: mwanza_crd_dashboard.png")

# Export CSV
if type_col:
    export = gdf[[type_col, 'geometry']].copy()
    export['longitude'] = export.geometry.x
    export['latitude'] = export.geometry.y
    export.drop(columns='geometry').to_csv('mwanza_businesses_stats.csv', index=False)
    print("Data exported: mwanza_businesses_stats.csv")
```

```
# Run the dashboard
cd ~/sdi-training
python3 crd_dashboard.py

# View the output image
xdg-open mwanza_crd_dashboard.png
```

**Checkpoint:**

API call returns list of Mwanza and Tanga datasets
GeoPandas loads Mwanza businesses with 100+ features
`mwanza_crd_dashboard.png` saved with map and chart
`mwanza_businesses_stats.csv` exported
You can adapt the script to load Tanga or Mwanza education data